

Purpose

The watchers sole purpose is to keep watch out for any new document that might arrive into the inbox directory and notify sentinel that the document is ready to be processed.

The watcher only keeps track if a new document has arrived in the inbox. It should not move, process, copy, classify, delete or decide where the document is supposed to go.

Responsibilities

- Monitor Configured Inbox Directory.
- Detect newly created files.
- Detect files moved into the inbox.
- Verify if a file is ready to be processed.
- Create a File Event.
- Send the file event to sentinel (main.py).
- Return to monitoring

Input

The watcher only receives

Inbox

For Example:

/home/hamza/Sentinel/inbox

Output

The Watcher generates a File event

Path:
/home/hamza/Sentinel/inbox/lecture5.pdf

Filename:
lecture.pdf

Extension:
PDF

Timestamp:
27-07-2026 11:30:55

Event:
Created

Hence the file event represents only the even of the file. Its name, extension, path, and timestamp. But does not know what the data inside the file, that will be a job for a different module namely the [Extractor](#).

The Algorithm:

We will follow a file readiness algorithm. This will ensure that the file event in only passed back to sentinel when the file has been fully pasted or created inside of the inbox. This will ensure to make the further work that sentinel does to be error free.

Explanation of the algorithm

1. Detect File creation
2. Record Current file size
3. Wait for a configurable interval.
4. Check file size again.
5. If the size changed:
 - Reset the stability timer.
 - Continue monitoring.
6. If the size remained unchanged:
 - Continue timing.
7. Once the file size has remained unchanged continuously for the configured stability period:
 - Consider the file ready.
8. Emit the File Event.

This algorithm is operating-system independent and avoids relying on platform-specific file locking behavior.

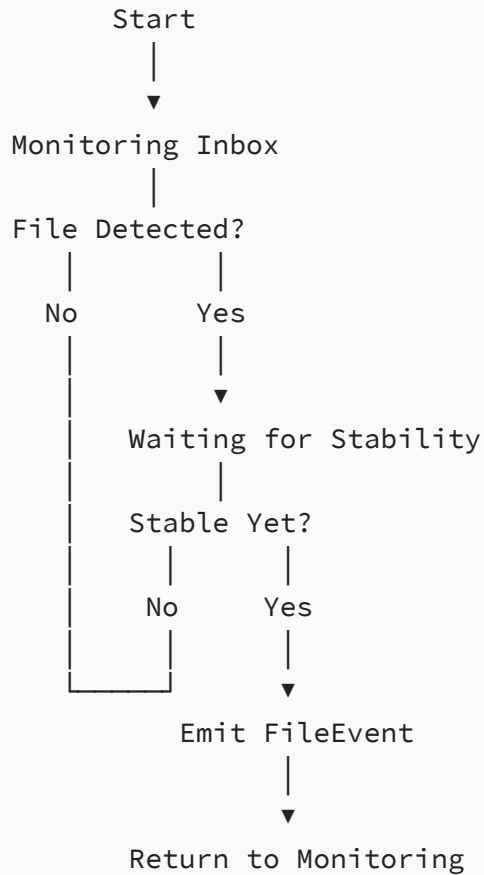
Logging Requirements

The Watcher shall produce informative, low-noise logs.

Examples:

```
[Watcher] Started.
```

State Diagram



Communication Contract

The watcher will never deal with any of the other modules of sentinel. It will only directly communicate with sentinel itself.

Tools required for the Implementation of the watcher:

Following tools will be used to implement the watcher into code:

- Python 3.11+
- pathlib (Cross Platform path handling regardless of Os).
- watchdog (Cross Platform file system monitoring).
- Data classes (For File Events).
- logging (library).
- typing (Intellisense).
- timing (used for the file readiness algorithm).

